Form 71 Mon. 24 May 2027 Weighted trees: multiplication along a path

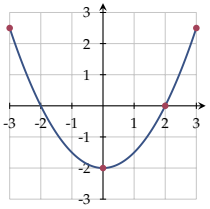
1. Warming · Flash Calculate $0.3 \times 0{,}9$.
2. Heart · Method With the tree of the sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.
3. Heart + · Method Deduct from it $P(T)$.
4. Return · Factorize Factoring $x^2 - 9$.
5. Red thread · Flash The side $[BC]$ of length 13 cm is hypotenuse of a right triangle ABC. What is the radius of his circumscribed circle?

Form 72 March 25 May 2027 From painting to tree and symbolic writing

1. Warming · Flash Express $\frac{30}{60}$ in the form of probability.
2. Heart · Method From the table on sheet 69, build the tree (F or Gthen S or $\overline{S}$) : give $P(F)$, $P_F(S)$ and $P_G(S)$.
3. Heart + · Translate Write symbolically and calculate "the probability of being a boy who does not make sport".
4. Back · Flash A car runs at constant speed: 150 km in 2 hours.How far in 5 hours?
5. Red thread · Statement⟶ formula A taxi charges 2,000 Ar of support and then 1,500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

Form 73 27 May 2027 Inversion of conditioning: P $_A(B)$ *et* $P_B(A)$

1. Warming · Flash What a difference between $P_M(T)$ and $P_T(M)$?
2. Heart · Method With data from form 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.
3. Heart + · Method Rare disease (1% of patients); positive test for 99% of patients and 5% of patients No malady. $P_T(M)$.
4. Back · Graph Curve of f :



Read $f(2)$ and the preimage of 2.5.

5. Red thread · Method Program: choose a number; multiply it by -2 ; add 10 ; divide by 2. Result for 7 ? Expression in x ?

Form 74 Mon. 31 May 2027 False positives and false negatives

1. Warming · Flash Define "false positive" and "false negative".
2. Heart · Method 10 000 people, 2 % sick; the test is positive for 90 % of patients and 10 % How many false negatives, false positives, positives in total?
3. Heart + · Interpret With the previous data, calculate the proportion of true patients among the positive and comment.
4. Return · True/false True or false? Justify. The area of a square is proportional to the length of sound Side.
5. Red thread · Refuter Conjecture: "for the integer n, $n^2 + n + 41$ is first." $n = 40$.

Consolidation, fluency and transition to the First

Form 75 Mar. 1 1st June 2027 Synthesis problems involving several areas

1. Warming · Flash Solve 3 $(x-1)=x+5$.

2. Heart · Method For what value of k points $A(1;1)$, $B(3;5)$ and $C(k;9)$ Are they aligned?

3. Heart + · Method A price increases by 10% and then decreases by 10%. What is the overall evolution?

4. Back · Flash Total area of a right pavement of dimensions 4, 3, 2 (cm) ?

5. Red thread · Flash Calculate the exact volume of a 3 cm radius ball.

Form 76 3 June 2027 Assessment of the six competences

1. Warming · Metacognition Citing the six mathematical skills worked in high school.

2. Heart · Metacognition Associate: "to translate a statement by an equation"; "to plot a curve"; "justify equality by a written calculation" to a competency.

3. Heart + · Metacognition To solve 2 $x+3=11$, which competence is mainly mobilized?

4. Back · Problem Water at 20 °C heats 3 °C per minute: $T(t)=20+3t$. After how many Does it reach 65 °C minutes?

5. Red thread · Flash Solve in $\mathbb{R}$: $0{,}5x+1{,}2=2{,}7$.

Form 77 Mon. 7 June 2027 Targeted Reactivation of Prerequisites for the First

1. Warming · Flash Develop $(x+3)^2$.

2. Heart · Method Verify that $x^2-5x+6=(x-2)(x-3)$ and deduce the solutions of $x^2-5x+6=0$.

3. Heart + · Anticipation For $f(x)=x^2$, calculate the rate of increase $\dfrac{f(3)-f(1)}{3-1}$.

4. Back · Construction programme We're setting M and N midpoints $[AB]$ and $[AC]$ And then we trace $[MN]$. That Can we say that (MN) and (BC) ? Calculate MN if $BC=9$ cm.

5. Red thread · Write Demonstrate that a triangle of sides 7, 24 and 25 is rectangle.

Form 78 March 8 June 2027 Synthesis issues: choosing and justifying a strategy

1. Warming · Code reading What's sent back $f(2)$?

```
def f(x):
    return x**2 - 3*x + 1
```

2. Heart · Completion Complete for function to return True where A, B, C are aligned:

```
def lines(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa)== ....
```

3. Heart + · Completion Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        Return x
    Else:
        return ....
```

4. Back · Flash Write in decimal notation 2.5 $\times 10^3$ then 6.2 $\times 10^{-2}$.

5. Red thread · Flash Give the symmetrical of $A(2;-3)$ in relation to origin.